MODULE 2 LOWER EXTREMITY

The Lower Extremity.

APPENDICULAR SKELETON

The pelvic girdle that carries the body's weight to the legs, and the bones of the thigh, leg, and foot, each broken out with its key markings.

BY THE END

1. Name the three bones that fuse to form the hip bone, and describe the pelvis.
2. Identify the key markings of the hip bone, including the acetabulum.
3. Identify the key markings of the femur, patella, tibia, and fibula.
4. Name the bones of the foot, including the seven tarsal bones and the arches.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Appendicular Skeleton, Lower

drsrennie-stack.github.io/new-build-bio4-solano/appendicular-concept-videos.html

The Lower Extremity, an Overview

The lower limb is built heavier than the upper limb because it carries the body's weight.

FOUR PARTS, PROXIMAL TO DISTAL

Part	Region it spans	Bones
Pelvic girdle	Attaches the lower limb to the axial skeleton	The two hip bones
Thigh	Hip to knee	Femur, and the patella at the knee
Leg	Knee to ankle	Tibia and fibula
Foot	Ankle to toes	Tarsals, metatarsals, and phalanges

The Pelvic Girdle

Each hip bone, the os coxae, begins as three separate bones that fuse at the acetabulum: the ilium, the ischium, and the pubis.

Ilium

MARKINGS

Marking	What it is
Ilium	The large, fan-shaped superior part of the hip bone
Iliac crest	The curved upper ridge, the rim you rest your hands on
Anterior superior iliac spine	The prominent point at the front of the crest, a palpable landmark
Posterior superior iliac spine	The point at the back of the iliac crest
Iliac fossa	The smooth, concave inner surface
Auricular surface	The ear-shaped surface that joins the sacrum at the sacroiliac joint
Greater sciatic notch	The deep notch on the posterior border. The sciatic nerve, the largest nerve in the body, passes through it

Ischium and pubis

MARKINGS

Marking	What it is
Ischium	The posteroinferior part of the hip bone
Ischial tuberosity	The strong, roughened knob you sit on
Ischial spine	The pointed projection above the tuberosity
Pubis	The anteromedial part of the hip bone
Pubic symphysis	The cartilage joint where the two pubic bones meet in front
Pubic crest	The ridge along the upper border of the pubis
Pubic rami	The superior and inferior bars of bone extending from the body

The acetabulum and the pelvis

WHERE THE THREE BONES MEET, AND THE PELVIS AS A WHOLE

Structure	What it is
Acetabulum	The deep cup where the three bones meet, the socket for the head of the femur
Obturator foramen	The large opening framed by the ischium and the pubis
The bony pelvis	The two hip bones together with the sacrum and the coccyx
True and false pelvis	The false pelvis is the wide flare above the pelvic brim. The true pelvis is the narrower ring below it
Sex differences	The female pelvis is wider and shallower, with a broader pubic angle and a larger outlet, adapted for childbirth

The Thigh and Knee

Femur

MARKINGS, PROXIMAL TO DISTAL

Marking	What it is
Femur	The thigh bone, the longest and strongest bone in the body
Head	The ball at the proximal end that fits into the acetabulum
Fovea capitis	The small pit on the head that anchors a ligament to the acetabulum
Neck	The narrow region just below the head, a common fracture site
Greater trochanter	The large lateral projection, a major muscle attachment
Lesser trochanter	The smaller projection on the medial side
Linea aspera	The rough ridge down the posterior shaft, a muscle attachment line
Medial and lateral condyles	The rounded distal knobs that articulate with the tibia
Medial and lateral epicondyles	The projections just above the condyles
Intercondylar fossa	The notch between the condyles on the posterior surface

Patella

THE SESAMOID AT THE KNEE

Feature	What it is
Patella	The kneecap, a sesamoid bone formed within the quadriceps tendon
Base	The broad superior border of the patella
Apex	The pointed inferior tip

The Leg

TIBIA AND FIBULA COMPARED

Bone	Side and weight	Proximal (knee) end	Distal (ankle) end
Tibia	Medial, the shin bone, the weight-bearing bone of the leg	Medial and lateral condyles articulate with the femur. Tibial tuberosity on the front anchors the patellar ligament. Sharp anterior border felt as the shin	Medial malleolus, the medial bump of the ankle
Fibula	Lateral, slender, does not bear body weight	Head, the knob at the proximal end	Lateral malleolus, the lateral bump of the ankle

KEY POINT

The tibia carries the weight and the fibula does not. That one fact tells you which malleolus is which, why a length of fibula can be harvested as a graft, and why a tibial fracture is the more serious of the two.

The Foot

Tarsals, the ankle

SEVEN BONES

Structure	What it is
Tarsals	Seven bones forming the ankle and the back of the foot
Talus	The tarsal that articulates with the tibia and fibula and receives the body's weight
Calcaneus	The heel bone, the largest tarsal
Other tarsals	The navicular, the cuboid, and the three cuneiform bones

Metatarsals, phalanges, and arches

THE SOLE AND THE DIGITS

Structure	What it is
Metatarsals	Five bones forming the sole, numbered 1 to 5 from the great toe
Phalanges	The toe bones, fourteen in each foot
Great toe	The hallux. Has two phalanges, the other toes have three each
Arches of the foot	The medial and lateral longitudinal arches and the transverse arch, which spread body weight

THE SKELETON THAT CARRIES YOUR WEIGHT

A broken hip is usually a fracture of the femoral neck, common in older adults whose bone has thinned with osteoporosis. The neck is narrow and angled, and the blood supply to the femoral head runs close by, so the fracture can threaten the head itself. The anterior superior iliac spine and the iliac crest are landmarks a clinician feels through the skin, used to locate injection sites and to check pelvic alignment. The tibia carries body weight and the fibula does not, which is why a length of fibula can be taken as a bone graft, and the lateral malleolus of the fibula is the common ankle fracture site, often from a rolled ankle. The arches of the foot spread weight across the sole and spring back with each step. When the medial longitudinal arch drops, the result is flat feet.